Lecture
Tuesday
28.04.2026
(to compensate
25 March 2026)

$\dot{x} = Ax, \quad A \in M_{2\times 2}(\mathbb{R}) \qquad A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$

$\rightsquigarrow P_2(\lambda) = \lambda^2 - \underbrace{\mathrm{tr}A}_{a+d}\lambda + \underbrace{\det A}_{ad-cb} = 0$

$\Delta := (\mathrm{tr}A)^2 - 4\det A$

(i) $\Delta > 0 \Rightarrow \exists\ \lambda_1, \lambda_2 \in \mathbb{R},\ \lambda_1 \neq \lambda_2,\ \lambda_{1,2} \in \sigma(A)$

(ii) $\Delta < 0 \Rightarrow \exists\ \lambda_1, \lambda_2 \in \mathbb{C},\ \lambda_1 = \bar{\lambda}_2,\ \lambda_{1,2} \in \sigma(A)$

(iii) $\Delta = 0 \Rightarrow \exists\ \lambda_1 = \lambda_2 \in \mathbb{R},\ \lambda_1 \in \sigma(A)$

GOAL: Study the cases (i)-(iii) with the aim of analysing (and explicating) the trajectories to see how the phase portrait looks like

TYPICAL STRATEGY: change of basis $\rightsquigarrow$ $A \sim$ diag
$\Rightarrow$ system is easier (and uncoupled)
$\Rightarrow$ expression of sol $\Rightarrow$ trajectories.

(i) $\lambda_{1,2} = [\dots]$

$\lambda_1 = \dfrac{\mathrm{tr}A + \sqrt{\Delta}}{2} \qquad \lambda_2 = \dfrac{\mathrm{tr}A - \sqrt{\Delta}}{2}$

$\rightsquigarrow (\lambda_1, w_1) \quad (\lambda_2, w_2) \rightsquigarrow \begin{cases} \dot{\xi} = \lambda_1 \xi \\ \dot{\eta} = \lambda_2 \eta \end{cases}$

* NOT TRIVIAL trajectories
* TRIVIAL $\rightsquigarrow$ nodes and saddle node (stable / unstable) depending on $\mathrm{sgn}(\lambda)$

(ii) $\lambda = \dfrac{\mathrm{tr}A}{2} + i\dfrac{\sqrt{-\Delta}}{2} \qquad \lambda = \dfrac{\mathrm{tr}A}{2} - i\dfrac{\sqrt{-\Delta}}{2}$

$\rightsquigarrow (\lambda, w) \quad (\bar{\lambda}, z), \quad w = u + iv$

$\lambda = \underbrace{\mu}_{\mathrm{Re}(\lambda)} + i\underbrace{\rho}_{\mathrm{Im}(\lambda)}$

Rmk:
1) $w = u + iv \Rightarrow u \perp v$
2) $(\lambda, w), (\bar{\lambda}, z) \Rightarrow z = \bar{w} = u - iv$

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⚠ $p = \eta$

while proving $z = \bar{w}$ ($Aw = \lambda w$ and $A\bar{w} = \bar{\lambda}\bar{w}$)

$\rightsquigarrow \begin{cases} Au = \mu u - \eta v \\ Av = \eta u + \mu v \end{cases}$ to distinguish the real and the imaginary part

By point 1), since $u \perp\!\!\!\perp v \Rightarrow$ we found a basis for $\mathbb{R}^2$

$\mathcal{B} = \{u, v\}$ and the representative matrix is $A = \begin{pmatrix} \mu & \eta \\ -\eta & \mu \end{pmatrix}$

with $\mu = \frac{\operatorname{tr} A}{2}$

and the sol writes as (see the general framework, 2 lectures ago) $p = \frac{\sqrt{-\Delta}}{2}$

(*) $\vec{x}(t) = \alpha w e^{\lambda t} + \beta \bar{w} e^{\bar{\lambda} t}$, $\alpha, \beta \in \mathbb{C}$ determined by IC.

claim

Rmk: since $\vec{x}(t) \in \mathbb{R}^2 \Longrightarrow \alpha = \bar{\beta}$

In order to do so you have to explicit (*) using

$\alpha = Re(\alpha) + i\, Im(\alpha)$

$\beta = Re(\beta) + i\, Im(\beta)$

$\lambda = \mu + ip$

$w = u + iv$

[...]

Then using that $\vec{x}(t) \in \mathbb{R}^2$ ! $\Rightarrow$ $\begin{cases} Re(\alpha) = Re(\beta) \\ Im(\alpha) = -Im(\beta) \end{cases}$

$\Updownarrow$

$\alpha = \bar{\beta}$.

In order to write the trajectories, some more effort is needed.

First from the Rmk, we write the sol as.

$$\vec{x}(t) = e^{\mu t}\left[2Re(\alpha)\left(u\cos(pt) - v\sin(pt)\right) - 2Im(\alpha)\left(u\sin(pt) + v\cos(pt)\right)\right]$$

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$\rightsquigarrow$ Imaginary part $= 0 \iff$

$$\big(Re(\alpha) - Re(\beta)\big)\big(u\sin(\mu t) + v\cos(\mu t)\big) - \big(Im(\alpha) + Im(\beta)\big)\big(u\cos(\mu t) - v\sin(\mu t)\big) = 0$$

$\alpha$ and $\beta$ are arbitrary

$\iff$ $\begin{cases} Re(\alpha) = Re(\beta) \\ Im(\alpha) = -Im(\beta) \end{cases}$

$\iff \alpha = \overline{\beta}$. ☒

Therefore, $x(t)$ rewrites as

$$x(t) = e^{\mu t}\left[2Re(\alpha)\left(u\cos(\mu t) - v\sin(\mu t)\right) - 2Im(\alpha)\left(u\sin(\mu t) + v\cos(\mu t)\right)\right]$$

(goniometric)

We now use the exponential form of $\alpha = \frac{\rho}{2} e^{i\theta} = \frac{\rho}{2}(\cos\theta + i\sin\theta)$ (*)

$\Downarrow$

$Re(\alpha) = \frac{\rho}{2}\cos\theta$

$Im(\alpha) = \frac{\rho}{2}\sin\theta$

$$x(t) = e^{\mu t}\left[2iu\left(Re(\alpha)\cos(\mu t) - Im(\alpha)\sin(\mu t)\right) - 2v\left(Re(\alpha)\sin(\mu t) + Im(\alpha)\cos(\mu t)\right)\right]$$

$$\overset{(*)}{=} \rho e^{\mu t}\left[u\left(\cos\theta\cos(\mu t) - \sin\theta\sin(\mu t)\right) - v\left(\cos\theta\sin(\mu t) + \sin\theta\cos(\mu t)\right)\right]$$

$$= \boxed{\rho e^{\mu t}\left[u\cos(\mu t + \theta) - v\sin(\mu t + \theta)\right]}$$

with $x(0) = 2u\,Re(\alpha) - 2v\,Im(\alpha) \Rightarrow \begin{cases} x(0) = 2Re\,\alpha u_x - 2Im\,\alpha v_x \\ y(0) = 2Re\,\alpha u_y - 2Im\,\alpha v_y \end{cases}$

To conclude, as far as $\alpha$ is known, you have got the explicit solution.

**GOAL** qualitative analysis,

with the basis $B = \{u, v\}$ $\rightsquigarrow$ $\begin{pmatrix} \dot{\xi} \\ \dot{\sigma} \end{pmatrix} = \begin{pmatrix} \mu & \eta \\ -\eta & \mu \end{pmatrix} \begin{pmatrix} \xi \\ \sigma \end{pmatrix}$

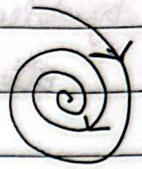
$\iff \begin{cases} \dot{\xi} = \mu\xi + \eta\sigma \\ \dot{\sigma} = -\eta\xi + \mu\sigma \end{cases} \Rightarrow \begin{cases} \xi(t) = \rho e^{\mu t}\cos(\mu t + \theta) \\ \sigma(t) = -\rho e^{\mu t}\sin(\mu t + \theta) \end{cases}$

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CASES: **1 CASE**: $\mu \neq 0$ : the rotation with angular velocity $\eta$ is modulated by the exponential fnct $e^{\mu t}$

* $\mu < 0 \Rightarrow x(t) \to 0 \quad t \to +\infty$

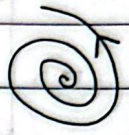
(unstable in the past) → stable spiral (asymptotically)



* $\mu > 0 \Rightarrow x(t) \to 0, \quad t \to -\infty$

(stable in the past) → unstable spiral

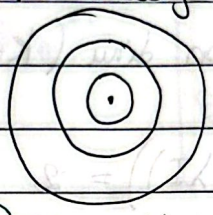
asymptot.



**2 CASE**: $\mu = 0$ : there are only rotations of angular velocity $\eta$

⚠ $Re(\lambda) = 0$ → the orbits are cicles → center

the steady state is stable but NOT asymp.



and to visualise better the oscillations

In order to study the orbits it's convenient to rewrite the system with polar coordinates

$\xi = r\cos\varphi \qquad \eta = r\sin\varphi$ ← it should be $\sigma$

$$\begin{pmatrix} \dot{\xi} \\ \dot{\eta} \end{pmatrix} = \begin{pmatrix} \cos\varphi & -r\sin\varphi \\ \sin\varphi & r\cos\varphi \end{pmatrix} \begin{pmatrix} \dot{r} \\ \dot{\varphi} \end{pmatrix}$$

by inverting

$$\begin{pmatrix} \dot{r} \\ \dot{\varphi} \end{pmatrix} = \frac{1}{r^{\star}} \begin{pmatrix} r\cos\varphi & r\sin\varphi \\ -\sin\varphi & \cos\varphi \end{pmatrix} \begin{pmatrix} \dot{\xi} \\ \dot{\eta} \end{pmatrix}$$

$$= \frac{1}{r} \begin{pmatrix} r\cos\varphi & r\sin\varphi \\ -\sin\varphi & \cos\varphi \end{pmatrix} \begin{pmatrix} \mu r\cos\varphi + \eta r\sin\varphi \\ -\eta r\cos\varphi + \mu r\sin\varphi \end{pmatrix}$$

RIC

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} ax+by \\ cx+dy \end{pmatrix}$$

$$= \frac{1}{r} \begin{bmatrix} \mu r^2\cos^2\varphi + \eta r^2\cos\varphi\sin\varphi - \eta r^2\cos\varphi\sin\varphi + \mu r^2\sin^2\varphi \\ -r\mu\cos\varphi\sin\varphi - \eta r\sin^2\varphi - \eta r\cos^2\varphi + \mu r\cos\varphi\sin\varphi \end{bmatrix}$$

$$= \begin{pmatrix} \mu r \\ -\eta \end{pmatrix} \Rightarrow \begin{cases} \dot{r} = \mu r \\ \dot{\varphi} = -\eta \end{cases}$$

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$$\Rightarrow \begin{cases} r(t) = r_0 e^{\mu t} \\ \theta(t) = \theta_0 - \eta t \end{cases}$$ ⇝ it's clear that we have a rotation with angular velocity $\eta$ + variation of the radius ($e^{\mu t}$) exponential

by deleating the time

$$\frac{r}{r_0} = e^{\mu t} \qquad t = \frac{1}{\mu}\log\left(\frac{r}{r_0}\right)$$

$$t = -\frac{1}{\eta}(\theta - \theta_0)$$

$$\Rightarrow -\frac{\mu}{\eta}(\theta - \theta_0) = \log\left(\frac{r}{r_0}\right)$$

$$\Rightarrow \boxed{r = r_0 e^{-\frac{\mu}{\eta}(\theta - \theta_0)}}$$

IF $\mu \neq 0 \Rightarrow$ logarithmic spiral.

(iii) $\Delta = 0 \quad \exists\, \lambda_1 = \lambda_2 = \frac{\operatorname{tr} A}{2}$ (with multiplicity 2)

The solution depends on $\dim(\ker(A - \lambda I))$

CASE 1 $\dim(\ker(A - \lambda I)) = 2$ (= max) ← all the space

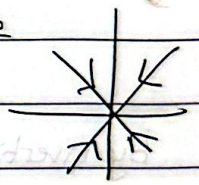
$\Rightarrow$ trivial case because $\forall\, w \in \mathbb{R}^2 \quad (A - \lambda I)w = 0$

i.e. $\forall$ vector is an eigenvector of $A$.

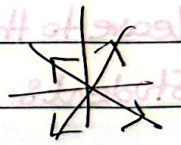
$\Rightarrow$ we can say that $A$ is already in the form $\Lambda = \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix}$ diagonal

↳ without any change of basis

$\Rightarrow x(t) = \alpha e^{\lambda t}$ ⇝ $\lambda < 0$ ⇝ star node stable



⇝ $\lambda > 0$ ⇝ unstable star node



CASE 2 $\dim(\ker(A - \lambda I)) = 1$

$\Rightarrow \exists!\, u \in \mathbb{R}^2 : \quad (A - \lambda I)u = 0$

In order to construct a basis for $\mathbb{R}^2$, we take another eigenvector $w$ s.t. $w \nparallel u \Rightarrow \{u, w\}$ basis for $\mathbb{R}^2$

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$\Rightarrow$ $Au = \lambda u$ because eigenvector of eigenvalue $\lambda$

$Aw = \alpha u + \mu w$

Therefore, the representation of $A$ in the basis $B = \{u, w\}$ is

$$\begin{pmatrix} \lambda & \alpha \\ 0 & \mu \end{pmatrix} \Rightarrow \lambda \text{ and } \mu \text{ eigenvalues}$$

$\Rightarrow \lambda = \mu$

↳ because $\exists! \lambda$

In order to write the normal form in this case, I choose $v = \frac{w}{\alpha}$ instead of $w \Rightarrow B = \{u, v\}$, $\begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix}$

In order to study the orbit, (and explicit the solution)

$$\begin{cases} \dot{\xi} = \lambda \xi + \sigma \\ \dot{\sigma} = \lambda \sigma \end{cases} \Rightarrow \begin{cases} \xi = \lambda \xi + \sigma_0 e^{\lambda t} \\ \sigma = \sigma_0 e^{\lambda t} \end{cases}$$

← non-homo. eq

by similarities $\xi(t) = \Gamma(t) e^{\lambda t}$

the Cauchy problem is totally solved

$$\begin{cases} \xi(t) = \sigma_0 t e^{\lambda t} + \xi_0 e^{\lambda t} \\ \sigma(t) = \sigma_0 e^{\lambda t} \end{cases} \quad (*)$$

GOAL: qualitative analysis

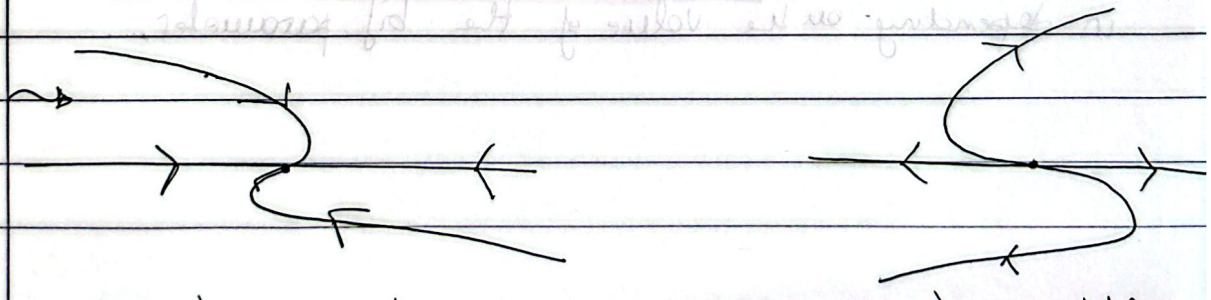
Pnt 1 $(\xi, 0)$ is invariant

For the non-trivial orbits we have from $(*)$

$$t = \frac{1}{\lambda} \log\left(\frac{\sigma}{\sigma_0}\right), \quad e^{\lambda t} = \frac{\sigma}{\sigma_0}$$

$$\xi = \frac{\sigma_0}{\lambda} \log\left(\frac{\sigma}{\sigma_0}\right) \frac{\sigma}{\sigma_0} + \xi_0 \frac{\sigma}{\sigma_0} = \frac{\sigma}{\lambda} \log\left(\frac{\sigma}{\sigma_0}\right) + \frac{\xi_0}{\sigma_0} \sigma$$

$$\Rightarrow \boxed{\frac{\xi}{\xi_0} = \frac{\sigma}{\sigma_0}\left(1 + \frac{\sigma_0}{\lambda \xi_0} \log\left(\frac{\sigma}{\sigma_0}\right)\right)}$$



$\lambda < 0$ stable degenerate node $\qquad$ $\lambda > 0$ unstable

CASE 3 : $\lambda = 0 \rightsquigarrow \Lambda = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$

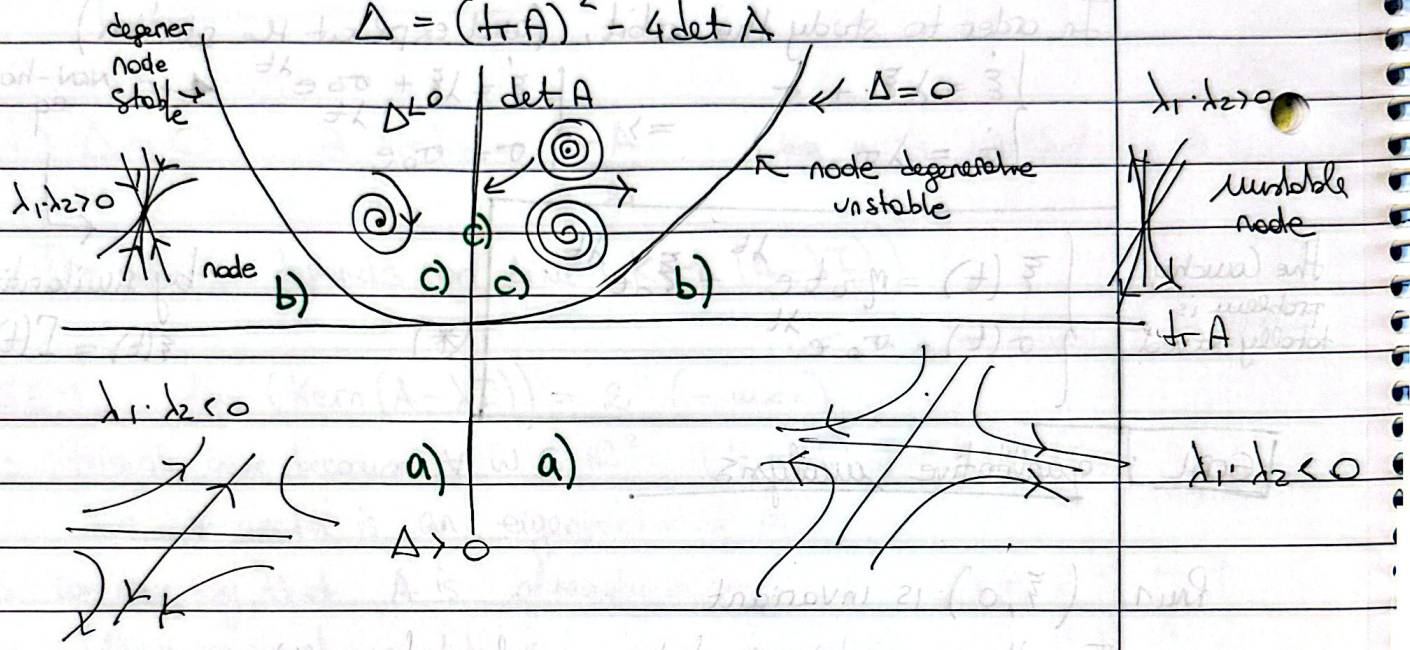
$$\Rightarrow \begin{cases} \dot{\xi} = \eta \\ \dot{\eta} = 0 \end{cases} \Rightarrow \begin{cases} \xi(t) = \eta_0 t + \xi_0 \\ \eta(t) = \eta_0 \end{cases}$$

$\Rightarrow$ // orbit at the axis $\xi$, at velocity $\eta$.

**CONCLUSIONS** : Linear diff. eq. in $\mathbb{R}^2$

$$\lambda^2 - \lambda \operatorname{tr} A + \det A = 0$$

$$\Delta = (\operatorname{tr} A)^2 - 4 \det A$$




a) $\lambda_1 \cdot \lambda_2 < 0$

b) $\lambda_1 \cdot \lambda_2 > 0$

c) $\lambda_1 \cdot \lambda_2 \in \not\subset$

The diagram is crucially usefull when your System depends on a parameter. Thus, you can understand how the orbits vary in depending on the value of the bif. parameter.

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